

الإجابة تكون في الجدول الموجود بأسفل صفحة الغلاف

1. If computed values are close to each others but far away from the true value, then we have;
 - A. high accuracy and high precision
 - B. low accuracy and high precision
 - C. low accuracy and low precision
 - D. high accuracy and low precision

2. The solution of the equation $ax^2 + bx + c = 0$ could be written as $x =$
 - A. $\frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$
 - B. $\frac{-2a}{b \pm \sqrt{b^2 - 4ac}}$
 - C. $\frac{2c}{b \pm \sqrt{b^2 - 4ac}}$
 - D. $\frac{2a}{b \pm \sqrt{b^2 - 4ac}}$

3. Use modified Newton-Raphson method to approximate the root of the equation; $x^3 - 2x^2 - 4x + 8 = 0$, if the initial guess $x_0 = 1$, then $x_2 =$
 - A. 1.789473684
 - B. 1.95959596
 - C. 1.993836672
 - D. 1.999791764

4. Approximate 12.478562149 to 4 significant digits, by Chopping ?
 - A. 12.47
 - B. 12.48
 - C. 12.4785
 - D. 12.4786

5. Normalize; 0.00000027×10^m , where m is an integer ?
 - A. $0.27 \times 10^{m+5}$
 - B. $0.27 \times 10^{m-5}$
 - C. $0.27 \times 10^{m+6}$
 - D. $0.27 \times 10^{m-6}$

6. Given that the equation $f(x) = 0$, has a root in the interval $[2, 2.5]$. Applying the bisection method to find this root, if the tolerance equals 0.00001, then the least number of iterations you should do is:
 - A. 14
 - B. 13
 - C. 16
 - D. 15

7. Given that 1 is a root of the equation; $x^5 - 3x^4 + 3x^3 - x^2 = 0$
then the multiplicity of this root equals:
 - A. 1
 - B. 3
 - C. 2
 - D. 4

8. Use bisection method twice to approximate the root of

$$\frac{20}{3+x} = 3 + \sqrt{x}, x \in [1, 2]$$

- A. 1.25 B. 0.75 C. 1.75 D. 0.25

9. Rewriting the equation;

$$x^3 + 4x - 4 = 0 \text{ in the form } x = a + bx^3 \text{ then the ordered pair } (a, b) =$$

- A. $(-\frac{1}{4}, 1)$ B. $(-\frac{1}{4}, 1)$ C. $(1, \frac{1}{4})$ D. $(1, -\frac{1}{4})$

10. Given that $x_{i+1} = \frac{1}{5} \sec^2 x_i$, $x_0 = 0.5$.

Find x_4 ? (Round your answer to 4 decimal places)

- A. 0.1716 B. 0.2090 C. 0.2091 D. 0.1717

11. Use three non-zero terms of Maclaurin series corresponding to the function $f(x) = \frac{1}{1+x^4}$

to approximate the value of the integral $\int_0^{0.6} \frac{dx}{1+x^4}$? (Round your answer to 6 decimal places)

- A. 0.585476 B. 0.746145 C. 0.749377 D. 0.585568

12. If the estimated height of a tower is 46 m, if it's true length is 50 m then the percent relative true error equals:

- A. 8% B. 12% C. 10% D. 14%

- 3) The volume of a cone (مخروط) is 10cm^3 , if its height; $h = 6\text{ cm}$. Use Newton-Raphson method twice to approximate its radius, take the initial guess to be 1. "Round off your answer to six decimal places"

Hint: the volume of the cone formula ($V = \frac{1}{3}\pi r^2 h$)

(4 Points)

$$V = 10$$

$$10 = \frac{1}{3}\pi r^2 h$$

$$10 = \frac{1}{3}\pi r^2 (6) \Rightarrow 0 = 2\pi r^2 - 10$$

$$\frac{30}{6\pi} = r^2$$

$$\sqrt{\frac{5}{\pi}} = r$$

$$1.261566 = r$$

true radius

الحقيقي

$$f(r) = 2\pi r^2 - 10$$

$$f'(r) = 4\pi r$$

$$r_1 = r_0 - \frac{2\pi r^2 - 10}{4\pi r}$$

$$r_0 = 1$$

$$r_1 = 1.295774715$$

$$r_2 = 1.262017813$$

$$r_2 \approx 1.262018$$

6 d.p

4